Manvansh Singh

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LinkedIn | GitHub | Portfolio

Education

Chandigarh Engineering College, CGC Landran — Landran, Punjab
B.Tech in Robotics & Artificial Intelligence — 07/2024 – Present

Technical Skills

Languages: Python, C/C++, JavaScript
Machine Learning: Regression, Classification (Random Forest, XGBoost), Anomaly Detection (Isolation Forest), Feature Engineering, Hyperparameter Tuning (GridSearchCV)
Deep Learning: PyTorch, CNNs, U-Net, Computer Vision
Frontend: HTML, CSS, React.js, Tailwind CSS
Backend & Database: Node.js, Express.js, FastAPI, MongoDB, MySQL, PostgreSQL
Python Libraries: NumPy, Pandas, Matplotlib, Seaborn, PyTorch, Scikit-learn, NiBabel, Plotly
Developer Tools: Git, GitHub, Postman, VS Code, Jupyter Lab, Vercel, Render

Experience

Devantra Innovations Pvt. Ltd. — Mohali
Software Developer — Jul 2026 – Present

- Led a team of 3 developers to design and deliver a full-stack Recruitment Drive Portal, building dedicated Admin and Candidate portals to automate the complete recruitment lifecycle.
- Engineered 26 REST API endpoints using Node.js, Express.js, and PostgreSQL for authentication, recruitment drives, candidate onboarding, assessments, reporting, and result management.
- Developed 8+ candidate-facing pages and 7 reusable React components, implementing secure role-based workflows and seamless frontend-backend integration.
- Designed, implemented, and deployed production-ready features while collaborating with cross-functional teams using Git/GitHub across multiple business applications.

Devantra Innovations Pvt. Ltd. — Mohali
Software Developer Intern — Jan 2026 – Jul 2026

- Architected and delivered 5 production ERP modules by engineering 18 REST API endpoints using PHP, Node.js, and PostgreSQL, supporting daily operations for **20+ employees and 5 administrators**.
- Implemented role-based authorization for Admin, Manager, and Member roles, enabling secure enterprise workflows across the ERP platform.
- Designed and integrated responsive interfaces across 10+ ERP pages, ensuring consistent user experience and seamless frontend-backend communication.
- Enhanced ERP Dashboard, Team Management, and workflow automation features while collaborating with the engineering team using Git/GitHub.

Projects

Turbofan Predictive Maintenance System | *Python, FastAPI, XGBoost, Scikit-learn, React* — GitHub

- Engineered 46 time-series features (rolling stats, RUL labels) from NASA's C-MAPSS dataset (100 engines, 20K+ cycles) with engine-level splitting to prevent temporal data leakage
- Trained RF/XGBoost models (GridSearchCV) achieving R^2=0.79 (RUL), F1=0.87, Recall=0.93; Isolation Forest flagged failures 12 cycles out vs. 113 for normal readings

Dental CBCT Tooth Segmentation Pipeline | *PyTorch, U-Net, Medical Imaging* — GitHub

- Built a full preprocessing-to-postprocessing 2D U-Net pipeline (7.7M params) for 3D CBCT tooth segmentation, trained on 14,790 slices from 50 volumes on RTX 4050 (CUDA)
- Achieved 0.7479 Dice / 0.6948 IoU on held-out test set with custom PyTorch DataLoader and interactive 3D Plotly visualization of predictions

AI-Powered Interview Preparation Platform | *MERN, JWT Auth, Gemini API* — GitHub

- Built an AI platform generating 8-10 role/experience-tailored interview questions via Gemini API, with "load more" (+4-5) and a pin/bookmark revision feature
- Enabled JWT authentication with protected routes and session expiry; deployed on Vercel and Render

Achievements

100+ LeetCode Problems Solved – consistent DSA practice across arrays, strings, hashing, and binary search
Finalist – Smart India Hackathon (SIH), Internal Round, 2025
1st Prize – CodeQuest (College-level Coding Competition)